

Dr. Shanu Dengre

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Summary

Senior Data Scientist with 4.5+ years of building statistical and ML models (time series, multivariate regression, XGBoost, causal inference, deep learning) and the training and inference pipelines that run them at scale, over millions of records in 10+ chains. Sets modelling best practices; mentors juniors. PhD in Physics.

Skills

ML Systems & MLOps: MLflow, Automated Retraining, Drift Monitoring, FastAPI, Docker, Kubernetes

Deep Learning: PyTorch, TensorFlow/Keras, Scikit-learn, CNNs, LSTM/GRU, Transformers

NLP & LLMs: NLP, Embeddings, LLM APIs, MCP

Causal Inference & Experimentation: Causal Inference/Uplift, A/B Testing, Statistical Inference

Statistical Modeling: Multivariate Regression, Time Series (ETS, ARIMA, Prophet), XGBoost/LightGBM

Data at Scale: Python, SQL, PySpark, Snowflake, MongoDB, AWS (SageMaker, S3, Lambda)

Forecasting & Vision: DARTS, Nixtla, CHRONOS, YOLO-family, OpenCV, SAM

Languages: English (Fluent), German (A2.2), Hindi (Native)

Experience

InterProductTec – Mumbai

Nov 2025 – Present

Sr. Data Scientist

Demand Forecast Project Lead

- Lead the team's modelling approach and data science best practices across concurrent customer projects.
- Innovated forecasting models (feature engineering, LightGBM/XGBoost, deep learning) improving sales forecast by 20% during holiday/offer seasons.
- Built hierarchical best-fit forecasting (ETS, ARIMA, Prophet, ML ensembles), tracking bias and MAPE.
- Designed backtests and A/B experiments to measure forecast and ordering impact on KPIs.
- Built POS data pipelines (Python, PySpark, SQL, Snowflake) processing millions of records.
- Built LLM-based agentic workflows to auto-link restaurant recipes to POS items.
- Built an MCP-based forecasting tool for drift detection that alerts when system accuracy drifts.
- Influence product, engineering and customer teams via results and trade-offs; mentor juniors.

PreciTaste – Munich

Feb 2022 – Oct 2025

AI Engineer and Data Scientist

Time Series Forecasting – Sales and Demand

- Led AI-driven demand forecasting across 10+ restaurant chains, reducing food waste by 15% and improving crew efficiency by 20%.
- Automated the DARTS training/retraining pipeline, cutting retraining time by 50%.
- Reduced query latency by 60% using FastAPI microservices with MongoDB, Docker, and Kubernetes.

Computer Vision – Object Detection and Segmentation

- Trained CNN food recognition models (YOLO-family, PyTorch) with MLflow, deployed across 5+ restaurant chains.
- Built automated annotation pipelines with OpenCV and SAM, reducing labeling time by 50%.
- Delivered a POS order-matching system with 95%+ real-time validation accuracy.

Live Demos

Next Restaurant Recommender | 🔗 next-restaurant.onrender.com

NLP, Embeddings

Newsvendor Problem Visualiser | 🔗 newsvendorproblem.streamlit.app

Python, Simulation

Education

PhD (Physics), TU Dresden | Publication: Phys. Rev. B 103, 064425 (2021)

Oct 2015 – Feb 2021

BSc–MSc (Physics), IISER Pune | Publication: Phys. Rev. B 95, 054401 (2017)

Aug 2010 – May 2015

Data Science & Machine Learning Bootcamp, Le Wagon Berlin

June 2021 – Sept 2021